

Aspectual Counting Framework (ACF): Aspects, Contexts, Bridges, and Higher-Order Conflict Geometry

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Abstract

The Aspectual Counting Framework (ACF) formalizes the idea that numerical answers depend on the *aspect* under which we count. An aspect fixes (i) admissible units, (ii) invariances, and (iii) permissible inferences within and across aspects via explicit *bridges*. This article develops ACF from a set-based quotient model through a functorial formulation, introduces contexts as a semilattice of practices, defines a bridge-selection principle as a prioritized satisfiable-subset problem, and analyzes the resulting conflict geometry via graphs, hypergraphs, Helly-type conditions, and a k -shadow hierarchy. We provide a concrete inconsistency theorem for Ship-of-Theseus-style paradoxes (as a non-existence of a global identity equivalence satisfying incompatible bridges), show how selection reduces to maximum-weight independent set (graph case) or its hypergraph analogue (higher-order case), define a practical Helly order induced by weights, give local dominance and antidominance criteria, and present a sound hypergraph-native certification procedure with a provable runtime-bounded local candidate generator. The only heuristic component is which candidate sets are tested; every certified constraint is sound.

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1 Motivation: aspectual answers and Theseus

In counting disputes (Ship of Theseus, restored artifacts, legal identity, software versions), parties often share all base-level facts yet disagree numerically. ACF posits: such disagreements frequently arise from *unlicensed cross-aspect inference*. One can count planks, ship-continuants, replacement events, legal registrations, functional roles, etc.; these are different invariants of the same underlying situation.

2 ACF-1: units as quotients and invariance

2.1 Situations and aspects (quotient form)

Let E be a domain of items (tokens, time-slices, events, etc.). Let Obj be a class of situations X with carriers $|X| \subseteq E$.

Definition 2.1 (Aspect as unitization). An *aspect* α assigns to each $X \in \text{Obj}$:

- (i) a selection $S_\alpha(X) \subseteq |X|$ (eligible candidates),
- (ii) an equivalence relation $\sim_{\alpha, X}$ on $S_\alpha(X)$ (when candidates count as the same unit).

Define the set of α -units $U_\alpha(X) := S_\alpha(X) / \sim_{\alpha, X}$.

Definition 2.2 (Aspectual count). If $U_\alpha(X)$ is finite, define

$$a(X, \alpha) := |U_\alpha(X)| \in \mathbb{N}.$$

2.2 Invariance by construction

Definition 2.3 (α -preserving bijection). A bijection $f : |X| \rightarrow |Y|$ is α -preserving if

- (i) $x \in S_\alpha(X) \iff f(x) \in S_\alpha(Y)$,
- (ii) $x \sim_{\alpha, X} x' \iff f(x) \sim_{\alpha, Y} f(x')$.

Theorem 2.1 (Invariance). If f is α -preserving, then $a(X, \alpha) = a(Y, \alpha)$.

2.3 Locality and additivity

Axiom 2.1 (Restriction coherence). If $Y \subseteq |X|$ and $X \upharpoonright Y$ is defined, then $S_\alpha(X \upharpoonright Y) = S_\alpha(X) \cap Y$ and $\sim_{\alpha, X \upharpoonright Y}$ is the restriction of $\sim_{\alpha, X}$.

Definition 2.4 (α -separation). If a disjoint sum $X \sqcup Y$ is defined, say $X \perp_\alpha Y$ if $U_\alpha(X \sqcup Y) \cong U_\alpha(X) \sqcup U_\alpha(Y)$.

Axiom 2.2 (Conditional additivity). If $X \perp_\alpha Y$ then $a(X \sqcup Y, \alpha) = a(X, \alpha) + a(Y, \alpha)$.

Remark 2.1. Continuant aspects often violate additivity over temporal concatenation: two history segments each contain one continuant, yet their concatenation also contains one.

3 ACF-2: functorial aspects, contexts, and bridges

3.1 Aspects as functors

Let $\mathcal{O}$ be a category of situations (objects are situations, morphisms are structure-preserving maps; isomorphisms are renamings).

Definition 3.1 (Aspect functor). An aspect is a functor $U_\alpha : \mathcal{O} \rightarrow \mathbf{FinSet}$. Define $a(X, \alpha) := |U_\alpha(X)|$ when finite.

Theorem 3.1 (Functorial invariance). If $g : X \cong Y$ in $\mathcal{O}$, then $U_\alpha(g) : U_\alpha(X) \cong U_\alpha(Y)$ and $a(X, \alpha) = a(Y, \alpha)$.

Remark 3.1. The quotient model of Definition 2.1 is recovered by taking $U_\alpha(X) = S_\alpha(X) / \sim_{\alpha, X}$.

3.2 Contexts as admissibility constraints

A key philosophical point is that admissibility depends on what a practice can *talk about*.

Definition 3.2 (Context-indexed situation categories). A context c supplies a category $\mathcal{O}_c$ of situations described in a language/signature L_c , and a forgetful/translation functor $F_c : \mathcal{O} \rightarrow \mathcal{O}_c$.

Definition 3.3 (Context-admissible aspect). An aspect is *admissible in context c* if there exists a functor $\bar{U}_{\alpha,c} : \mathcal{O}_c \rightarrow \mathbf{FinSet}$ such that $U_\alpha \cong \bar{U}_{\alpha,c} \circ F_c$.

3.3 Refinement as natural surjection

Definition 3.4 (Refinement). An aspect β *refines* α (write $\beta \succeq \alpha$) if there exists a natural transformation $\rho : U_\beta \Rightarrow U_\alpha$ such that each component $\rho_X : U_\beta(X) \rightarrow U_\alpha(X)$ is surjective.

Proposition 3.1 (Monotonicity under refinement). *If $\beta \succeq \alpha$ and both counts are finite, then $a(X, \beta) \geq a(X, \alpha)$.*

3.4 Bridges as natural relations

Definition 3.5 (Bridge as natural relation). Fix a context c . A bridge $\tau : U_\alpha \rightsquigarrow U_\beta$ is a natural family of relations $\tau_X \subseteq U_\alpha(X) \times U_\beta(X)$.

Remark 3.2. Bridges are often partial, context-dependent, and relational rather than functional. ACF enforces: *no bridge, no transfer*.

4 Ship of Theseus as non-existence of a global identity

4.1 A minimal inconsistency model

Let $S = \{s_0, s_1, s_2\}$ be three ship-slices: original at t_0 , maintained at t_1 , reassembled at t_1 . Let $\# \subseteq S^2$ encode co-present distinctness: $s_1 \# s_2$. Let C (continuity) relate s_0 to s_1 , and M (material provenance) relate s_0 to s_2 .

Theorem 4.1 (Theseus inconsistency theorem). *There is no equivalence relation $\sim$ on S such that: (i) $C(x, y) \Rightarrow x \sim y$, (ii) $M(x, y) \Rightarrow x \sim y$, (iii) $x \# y \Rightarrow \neg(x \sim y)$.*

Proof. From (i) and $C(s_0, s_1)$, $s_0 \sim s_1$. From (ii) and $M(s_0, s_2)$, $s_0 \sim s_2$. Transitivity yields $s_1 \sim s_2$, contradicting (iii) since $s_1 \# s_2$. $\square$

Remark 4.1. The “paradox” is the implicit demand that one identity relation satisfy incompatible bridge-induced constraints.

5 ACF-3: contexts as a semilattice and bridge selection

5.1 Contexts as a join-semilattice

Definition 5.1 (Context semilattice). Let $(\mathcal{C}, \leq)$ be a poset of contexts. Assume a join operation $c \vee d$ (least context extending both). Intuitively: $c \vee d$ combines practices, constraints, and bridge libraries.

5.2 #-strict contexts and satisfiability

Fix $c \in \mathcal{C}$. Let B_c be available bridge schemata. Each schema $b \in B_c$ induces for each situation X a required-identification relation $R_{b,c}(X)$ on candidate slices $S_c(X)$.

Define forced identity for $\Gamma \subseteq B_c$ on X by

$$\equiv_{\Gamma,c}(X) := \text{EqCl}\left(\bigcup_{b \in \Gamma} R_{b,c}(X)\right).$$

Let $\#_c(X) \subseteq S_c(X)^2$ be co-presence distinctness.

Definition 5.2 (#-strictness and satisfiability). Context c is *#-strict* if co-presence distinctness is non-negotiable. A set $\Gamma \subseteq B_c$ is *satisfiable* in c if for all X ,

$$\equiv_{\Gamma,c}(X) \cap \#_c(X) = \emptyset.$$

5.3 Prioritized satisfiable-subset selection

Let $\preceq_c$ be a priority preorder on B_c , inducing levels $L_1, \dots, L_K$ (highest to lowest). Encode lexicographic priorities by weights $w(b) = M^{K-i}$ for $b \in L_i$, $M > |B_c|$.

Definition 5.3 (Selection problem). Choose $\Gamma \subseteq B_c$ maximizing $\sum_{b \in \Gamma} w(b)$ subject to Γ satisfiable.

6 Conflict geometry: from graphs to hypergraphs

6.1 Monotonicity and minimal unsatisfiable sets

Lemma 6.1 (Monotonicity). If $\Gamma \subseteq \Delta$ and $\text{Unsat}(\Gamma)$ then $\text{Unsat}(\Delta)$.

Definition 6.1 (Minimal unsatisfiable sets (MUS)). A set $M \subseteq B_c$ is a MUS if $\text{Unsat}(M)$ and every proper subset is satisfiable. Let $\mathcal{M}_c$ denote the family of all MUS.

Lemma 6.2 (MUS basis). $\text{Unsat}(\Gamma)$ iff $\exists M \in \mathcal{M}_c$ with $M \subseteq \Gamma$.

6.2 Graph reduction axiom (BW2 / MUS2)

Axiom 6.1 (MUS2). All minimal unsatisfiable sets have size 2: $\forall M \in \mathcal{M}_c, |M| = 2$.

Under MUS2, satisfiable sets are exactly independent sets in a graph.

6.3 Conflict hypergraph (general case)

Definition 6.2 (Conflict hypergraph). Define $H_c = (V_c, \mathcal{E}_c)$ with $V_c = B_c$ and $\mathcal{E}_c = \mathcal{M}_c$. A bridge set Γ is satisfiable iff it contains no hyperedge: $\forall e \in \mathcal{E}_c, e \not\subseteq \Gamma$.

6.4 Reduction to (hyper)graph independent set

Proposition 6.1 (Optimization form). Bridge selection is equivalent to:

$$\max \left\{ \sum_{b \in \Gamma} w(b) : \Gamma \subseteq B_c, \forall M \in \mathcal{M}_c, \Gamma \not\supseteq M \right\}.$$

As an ILP:

$$\max \sum_b w(b)x_b \quad s.t. \quad \sum_{b \in M} x_b \leq |M| - 1 \quad (\forall M \in \mathcal{M}_c), \quad x_b \in \{0, 1\}.$$

7 Helly-type characterization and the k -shadow hierarchy

7.1 Helly number of satisfiability

Definition 7.1 (Helly number). Let $h(c) := \max\{|M| : M \in \mathcal{M}_c\} \in \mathbb{N} \cup \{\infty\}$.

Theorem 7.1 (Helly/MUS equivalence). *For $k \geq 2$, the following are equivalent: (i) every MUS has size $\leq k$; (ii) satisfiability is k -Helly (if all subsets of size $\leq k$ are satisfiable then the whole set is satisfiable); (iii) $\text{Unsat}(\Gamma) \Rightarrow \exists \Delta \subseteq \Gamma$ with $|\Delta| \leq k$ and $\text{Unsat}(\Delta)$.*

Corollary 7.1 (Graph reduction). *The conflict hypergraph reduces to a graph iff $h(c) = 2$ (equivalently MUS2).*

7.2 k -shadow approximation

Definition 7.2 (k -shadow of MUS family). For a MUS M and $k \geq 2$, define

$$\partial_k(M) := \begin{cases} \{S \subseteq M : |S| = k\}, & |M| > k, \\ \{M\}, & |M| \leq k. \end{cases} \quad \partial_k \mathcal{M}_c := \bigcup_{M \in \mathcal{M}_c} \partial_k(M).$$

Define the k -shadow approximation hypergraph $H^{(k)} = (B_c, \partial_k \mathcal{M}_c)$.

Proposition 7.1 (Equivalent per-MUS cap). Γ is feasible in $H^{(k)}$ iff for all $M \in \mathcal{M}_c$,

$$|\Gamma \cap M| \leq \min(k - 1, |M| - 1).$$

Theorem 7.2 (Monotone interpolation). *Let OPT_k be the maximum weight over Feas_k (the feasible sets of $H^{(k)}$), and $OPT_\star$ the true optimum. Then*

$$OPT_2 \leq OPT_3 \leq \dots \leq OPT_{h(c)} = OPT_\star.$$

8 Practical Helly order and early exactness under weights

8.1 Practical Helly order

Definition 8.1 (Practical Helly order). Define

$$h_w(c) := \min\{k \geq 2 : OPT_k = OPT_\star\}.$$

Definition 8.2 (MUS-load). For $\Gamma \subseteq B_c$, define $\text{load}(\Gamma) := \max_{M \in \mathcal{M}_c} |\Gamma \cap M|$. Let

$$L := \min\{\text{load}(\Gamma) : \Gamma \in \arg \max_{\Delta \in \text{Feas}_\star} w(\Delta)\}.$$

Theorem 8.1 (Exactness criterion via optimal-load). *For $k \geq 2$,*

$$OPT_k = OPT_\star \iff k \geq L^{+1}.$$

Consequently $h_w(c) = L^{+1}$ and $2 \leq h_w(c) \leq h(c)$.

9 Local dominance: sufficient conditions for early exactness

9.1 Hypergraph-native dominance (MUS-incidence)

Let $\mathcal{M}(b) := \{M \in \mathcal{M}_c : b \in M\}$.

Definition 9.1 (Incidence dominance). Define $d \succeq b$ if $w(d) \geq w(b)$ and $\mathcal{M}(d) \subseteq \mathcal{M}(b)$.

Lemma 9.1 (Safe swap). *If $d \succeq b$ and Γ is satisfiable with $b \in \Gamma$, then $(\Gamma \setminus \{b\}) \cup \{d\}$ is satisfiable and has weight no smaller.*

Definition 9.2 (k -tail). For each MUS M , let $Top_{k-1}(M)$ be the $\min(k-1, |M|)$ highest-weight elements of M . Define $Tail_k(M) := M \setminus Top_{k-1}(M)$ and $Tail_k := \bigcup_M Tail_k(M)$.

Assumption 9.1 (Local dominance condition $DC(k)$). For every $b \in Tail_k$, there exists $d \neq b$ with $d \succeq b$.

Theorem 9.1 (Local dominance implies $h_w(c) \leq k$). *If $DC(k)$ holds in a $\#$ -strict context, then $OPT_k = OPT_*$, hence $h_w(c) \leq k$.*

9.2 Graph-approx dominance (2-section neighborhood)

Define the 2-section graph G^{2sec} on B_c where $\{u, v\}$ is an edge iff they co-occur in some MUS. Let $N[b]$ denote closed neighborhood in G^{2sec} .

Definition 9.3 (Graph dominance). $d \succeq_G b$ if $w(d) \geq w(b)$ and $N[d] \subseteq N[b]$.

Assumption 9.2 (Graph dominance condition $DC_G(k)$). For every $b \in Tail_k$, there exists $d \neq b$ with $d \succeq_G b$.

Theorem 9.2 (Tail elimination for MWIS on G^{2sec}). *If $DC_G(k)$ holds, then G^{2sec} has a maximum-weight independent set that avoids $Tail_k$.*

10 Dual local lower bounds: ant domination structures

10.1 Local forcing of vertices (graph setting)

Let G be a weighted graph. For $b \in V(G)$, let $UB(b)$ be any upper bound on the maximum weight of an independent set in $G[N(b) \setminus \{b\}]$.

Lemma 10.1 (Local forcing). *If $w(b) > UB(b)$, then every MWIS contains b .*

10.2 Weakened packing via an auxiliary graph and matching

Fix k and tail set $T = Tail_k$ in G^{2sec} . For $u, v \in T$ nonadjacent, let $Z(u, v) = N[u] \cup N[v]$. If $w(u) + w(v)$ exceeds an upper bound on the best independent-set weight within $Z(u, v) \setminus \{u, v\}$, then every MWIS must include at least one of $\{u, v\}$.

Define an auxiliary graph A_k on vertex set T , with edge $\{u, v\}$ if the above certificate holds. Then any MWIS must hit each edge of a matching; hence matching size lower-bounds unavoidable tails.

Theorem 10.1 (Matching lower bound). *Let μ_k be the maximum matching size in A_k . Then every MWIS I satisfies $|I \cap Tail_k| \geq \mu_k$.*

11 Hypergraph-native certified unavoidable constraints and r -uniform packing

11.1 Certified unavoidable tail-hyperedges

Fix tail level k and choose $Q \subseteq \text{Tail}_k$ with $|Q| = r$.

Define

$$\mathcal{M}(Q) := \{M \in \mathcal{M}_c : M \cap Q \neq \emptyset\}, \quad Z(Q) := \bigcup_{M \in \mathcal{M}(Q)} (M \setminus Q), \quad \mathcal{M}_Z(Q) := \{M \in \mathcal{M}_c : M \subseteq Z(Q)\}.$$

11.2 Standard local bound: LP relaxation

Definition 11.1 (Local LP upper bound). Define $UB_{LP}(Q)$ as the optimum of

$$\begin{aligned} \max \quad & \sum_{b \in Z(Q)} w(b)x_b \\ \text{s.t.} \quad & \sum_{b \in M} x_b \leq |M| - 1 \quad (\forall M \in \mathcal{M}_Z(Q)), \\ & 0 \leq x_b \leq 1 \quad (\forall b \in Z(Q)). \end{aligned}$$

Definition 11.2 (Certified unavoidable set). Assuming Q itself contains no MUS (i.e. $\forall M \in \mathcal{M}_c, M \not\subseteq Q$), call Q *certified unavoidable* if

$$w(Q) > UB_{LP}(Q).$$

Theorem 11.1 (Unavoidability soundness). *If Q is certified unavoidable, then every maximum-weight feasible selection Γ intersects Q .*

11.3 Auxiliary hypergraph and fractional matching lower bound

Let $\mathcal{E}_{k,r}$ be the family of certified unavoidable r -sets. Define $A_{k,r} = (\text{Tail}_k, \mathcal{E}_{k,r})$.

Theorem 11.2 (Fractional matching tail lower bound). *Let $\nu_f(A_{k,r})$ be the fractional matching number of $A_{k,r}$. Then for every optimal selection Γ ,*

$$|\Gamma \cap \text{Tail}_k| \geq \tau(A_{k,r}) \geq \nu_f(A_{k,r}),$$

where τ is the transversal number.

Remark 11.1. For $r = 2$, this recovers the graph/matching bound in polynomial time. For $r \geq 3$, integral hypergraph matching is hard in general, but ν_f remains polynomial via LP.

12 Algorithm specification and soundness guarantees

12.1 ACF certification procedure (compact)

Inputs. B , weights w , MUS family $\mathcal{M}_c$, tail rule Tail_k , target (k, r) values, and a candidate generator producing finite families $\mathcal{Q}_{k,r} \subseteq \binom{\text{Tail}_k}{r}$.

Outputs. Certified families $\mathcal{E}_{k,r} \subseteq \mathcal{Q}_{k,r}$, auxiliary hypergraphs $A_{k,r}$, and lower bounds $L_{k,r} := \nu_f(A_{k,r})$.

Certification. For each $Q \in \mathcal{Q}_{k,r}$: (i) discard if it contains a MUS; (ii) compute $Z(Q)$ and $\mathcal{M}_Z(Q)$; (iii) solve the local LP to get $UB_{LP}(Q)$; (iv) if $w(Q) > UB_{LP}(Q)$, add Q to $\mathcal{E}_{k,r}$.

Bounding. Compute $\nu_f(A_{k,r})$ via the fractional matching LP.

12.2 Soundness theorem (certified vs heuristic)

Theorem 12.1 (Procedure soundness). *Fix a #-strict context with MUS family $\mathcal{M}_c$ and weights w . Run the certification procedure with any candidate sets $\mathcal{Q}_{k,r}$ (possibly generated heuristically). Then:*

- (i) *Every certified $Q \in \mathcal{E}_{k,r}$ is a valid unavoidable constraint: $\forall \Gamma \in \text{OPT}, \Gamma \cap Q \neq \emptyset$.*
- (ii) *Consequently every optimum hits every hyperedge in $A_{k,r}$.*
- (iii) *The computed bound $L_{k,r} = \nu_f(A_{k,r})$ satisfies $\forall \Gamma \in \text{OPT}, |\Gamma \cap \text{Tail}_k| \geq L_{k,r}$.*
- (iv) *The only heuristic element is coverage: failing to test some Q can miss constraints (false negatives), but the procedure cannot certify a false unavoidable constraint (no false positives), assuming LPs are solved correctly.*

13 A minimal local candidate generator with runtime bound

13.1 MUS co-incidence neighborhood generator (MCN)

Fix k and $T = \text{Tail}_k$.

Co-incidence graph. Build $C_k = (T, E_k)$ with edge $\{u, v\}$ iff $\exists M \in \mathcal{M}_c$ such that $\{u, v\} \subseteq M$. This can be built in time $O(\sum_{M \in \mathcal{M}_c} |M \cap T|^2)$.

Caps. Choose a neighborhood cap $s \in \mathbb{N}$ and radius $\ell \in \{1, 2\}$. For each $t \in T$, let $N^{(\ell)}(t)$ be its ℓ -hop neighborhood in C_k . Define $N_s(t)$ as the top $\min(s, |N^{(\ell)}(t)|)$ vertices in $N^{(\ell)}(t)$ by weight.

Star enumeration. For each target size r and each $t \in T$, generate candidates

$$\mathcal{Q}_{k,r}(t) := \{\{t\} \cup S : S \subseteq N_s(t), |S| = r - 1\}.$$

Let $\mathcal{Q}_{k,r} = \bigcup_{t \in T} \mathcal{Q}_{k,r}(t)$ (deduplicated).

Proposition 13.1 (Generator runtime bound). *For fixed k , the MCN generator runs in*

$$O\left(\sum_{M \in \mathcal{M}_c} |M \cap T|^2\right) + O\left(|T| \sum_{r \in R} \binom{s}{r-1}\right)$$

time (plus linear hashing for deduplication).

Remark 13.1. Soundness remains unconditional (Theorem 12.1); the generator only affects completeness.

14 Empirical and computational predictions

14.1 Graph vs hypergraph as a testable hypothesis

The MUS size profile $p_k = \frac{|\{M \in \mathcal{M}_c : |M|=k\}|}{|\mathcal{M}_c|}$ and Helly number $h(c)$ quantify higher-order conflict. Domains with $h(c) = 2$ are effectively pairwise; domains with frequent $|M| \geq 3$ support “all pairs ok, some triples forbidden” phenomena.

14.2 k -shadow saturation and order of interaction

Compute OPT_k (or bounds thereof) across k to estimate $h_w(c)$. Compute certified lower bounds on unavoidable tails via $L_{k,r}$; growth of $L_{k,r}$ with r indicates higher-order interaction beyond pairs.

14.3 Default parameter heuristics

Typical regimes suggest starting with $r \in \{2, 3\}$ and modest neighborhood caps (e.g. $s \approx 15$) with $\ell = 1$, escalating to $\ell = 2$ when MUS are larger or co-incidence neighborhoods are sparse.

15 Conclusion and open directions

ACF transforms counting/identity disputes into explicit choices of (i) aspects, (ii) contexts, and (iii) bridges. The resulting selection problem has a rich geometry: graph-like under MUS2, hypergraph-like otherwise, with Helly numbers measuring interaction order. The k -shadow hierarchy interpolates between pairwise approximations and true constraints, while $h_w(c)$ captures the effective order needed under a weight regime. Local dominance and certified unavoidability procedures yield fast, interpretable sufficient conditions and lower bounds. Future work includes: learning $\mathcal{M}_c$ from data, context morphisms and joins as empirical shifts, and characterizing when higher-order structure is necessary for stable disagreement.